

Use-Case Flow Document

Municipal Solid Waste Route Optimizer | UC-02: Generate Optimized Collection Routes

Primary actor: Sanitation Supervisor

Goal: Generate optimized daily collection routes for bins that need to be collected.

System: Municipal Solid Waste Route Optimizer

Preconditions

Before the use case starts, the Sanitation Supervisor must be authorized to use the system, and current fill-level information from the smart trash bins should be available.

Postconditions

When the process finishes successfully, optimized collection routes have been generated. The routes can then be assigned to the available truck drivers.

Main Success Scenario

1. The **Sanitation Supervisor** selects the option to generate the daily collection routes.
2. The **system** retrieves the latest fill-level information from the smart trash bins.
3. The **system** checks that the received fill-level data is valid and complete.
4. The **system** identifies bins that have exceeded the **80% fill threshold** or have been manually tagged for collection.
5. The **system** calculates optimized collection routes for the available trucks.
6. The **system** displays the generated routes and the bins included in each route.
7. The **Sanitation Supervisor** reviews the proposed routes.
8. The **Sanitation Supervisor** confirms the routes and assigns them to truck drivers.
9. The **system** saves the assignments and makes each route available to its respective truck driver.
10. The use case ends successfully.

Alternate Flow

A1. Invalid or Missing Bin Data

- A1.1 The system detects missing or invalid fill-level data for one or more bins.
- A1.2 The system informs the Sanitation Supervisor which bins are affected.
- A1.3 The affected bins are excluded from automatic route generation unless the supervisor manually tags them for collection.
- A1.4 The system continues generating routes using the remaining valid bins and any manually tagged bins.
- A1.5 The process continues from Step 5 of the Main Success Scenario.

A2. No Bins Exceed the 80% Threshold

- A2.1 The system finds that no bins have exceeded the 80% fill threshold.
- A2.2 The system informs the Sanitation Supervisor.
- A2.3 The supervisor may still include manually tagged bins, if there are any.
- A2.4 If there are no bins requiring collection, the use case ends without generating a collection route.